

Peng Kuang

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About me

I'm a final-year Ph.D. student in Computer Science at Lund University. My research intersects **programming systems**, **eye tracking**, and **human-centered AI**. I study programmers' cognitive processes via gaze and co-design tools to enhance their productivity and experience.

Areas of specialization

Programming Languages & Tools • Empirical Software Engineering • Distributed Machine Learning

Coursework

Machine Learning

Artificial Intelligence & Machine Learning 2025 (ongoing)
Scalable Data Science & Distributed Machine Learning
Applied Machine Learning
Deep Learning
Knowledge Technologies

Computer Science

Compilers
Distributed Systems
Mobile Computing Systems
Advanced Database Systems
Information Retrieval Systems
Enterprise Web Application Development
Computer Architecture & Operating Systems
Data Communications
Autonomous Systems
Interaction Design and Usability
Programmer Experience
Centric Methods for Designing Languages and Tools
Innovation & Entrepreneurship in IT
Enterprise Systems
Software Engineering
Project Management

Biometrics

Eye tracking
Functional and structural brain imaging (MRI/fMRI/dMRI/PET)
Electroencephalogram (EEG/MEG)

Informally Audited

Program Analysis
Cluster and Cloud Computing
Web Search and Text Analysis
Introduction to Quantum Computing (Udemy)
IQM Quantum School

Research Interests

Multimodal Intelligence (for education & health) / HCI & Brain-Computer Interface / Quantum Computing (programming languages & tools) / Virtual & Extended Reality / AI for Software Engineering

RECENT APPOINTMENTS

2021–present	Ph.D. Candidate/Teaching Assistant (defense on May 13, 2026) DEPT. OF COMPUTER SCIENCE, LU · Lund, SE - Agile Software Development (Java & Linux/Bash, four semesters) - Software Testing (Java, two semesters) - Large-scale Software Development (Java & JS, two semesters) - Object-Oriented Modeling and Design (Java, two semesters) - Database Technology (SQL & Java, one semester) - Scientific Theories for Engineers (one semester)
2025 Spring	Visiting Ph.D. Student PEACH LAB, DEPT. OF COMPUTER SCIENCE, ETH ZURICH · Zurich, CH
2020–2021	Business System Developer GUANGDONG TECHNION-ISRAEL INSTITUTE OF TECHNOLOGY(GTIIT) · Shantou, CN - ERP system implementation monitoring (Oracle & Java), and Student Life-cycle Management system (SLCM) research and evaluation. - Integration of the printing system PaperCut with payment service providers, e.g., Alipay. - Migration of the open-source asset management system SnipeIT. - Support for the open-source online learning system Moodle.
2019 Summer	IT Project Officer (part-time) BUSINESS SERVICES, UNIMELB · Melbourne, AU - Tested new versions of the Research Hub website, updated the web pages for the project, and assisted the team in other areas such as data analysis, workflow automation and project management.
2018–2019	Research Assistant (part-time) SCHOOL OF COMPUTING AND INFORMATION SYSTEMS, UNIMELB · Melbourne, AU - Researched about the health-seeking behavior of a minority cohort of international students in Australia. - Secured a partnership with a leading social app for participant recruitment.
2017–2019	President & Councilor (part-time) GRADUATE STUDENT ASSOCIATION, UNIMELB · Melbourne, AU - Led the council to govern the operations of the organization, including strategic planning, financial management, risk management, and policies and procedures development.
2018 Spring	Web Developer (intern) PSYCHPRESS · Melbourne, AU - Implemented the mobile-friendly authentication features and fixed 5 issues (data-querying & loading speed) for existing web applications with PHP/Laravel.

EDUCATION

Master of Information Technology , <i>Distributed Computing</i> , University of Melbourne	2019 Dec
Graduate Certificate in IT , <i>WAM 81</i> , Monash University	2016 Dec
Bachelor of Literature , <i>Advertising</i> , Xiangtan University	2005 Jun

INDUSTRIAL EXPERIENCE

2015–2016	Operations Lead LEADON CO., LTD. · Shenzhen, CN
2014–2015	Co-founder APPCOOL CO., LTD. · Chengdu, CN
2013–2014	Game Monetization Manager GAMELOFT SE · Chengdu, CN
2012–2013	Product Manager ZYNGA INC. · Beijing, CN
2010–2012	Community Manager ZYNGA INC. · Beijing, CN
2008–2010	Product Marketing & Operation Manager LINEKONG INTERACTIVE · Beijing, CN
2006–2007	Marketing Specialist RUNDIAN GAMEWARE · Beijing, CN
2005–2006	Oversea Marketing Specialist SUNIDVR · Shenzhen, CN

Programming

Python/Java	
Linux/Shell	
Qiskit/Cirq/Q#/.Silq	
JS/PHP/C#/R	
HTML/CSS/SQL	

Tools

Git / Docker / Gradle / Jenkins
PyTorch / TensorFlow / React
Agile / Jira / Unity / Figma
ERP / CRM / Salesforce / SAP

Sensors & Datasets

time series / eye-tracking
(gaze) / code / natural
language text / ECG / MRI
(brain images)

Research Methods & Skills

Survey / Interview /
Workshop / Case Study /
Controlled Experiment /
Systematic Literature Review /
Quantitative / Qualitative /
Mixed

Certificates

IQM Quantum School
AWS Cloud Practitioner
Essentials
Google Cloud Platform
Fundamentals: Core
Infrastructure
Responsive Web Design
JavaScript Algorithms & Data
Structures

Languages

Chinese									
English	●	●	●	●	●	●	●	●	●
Swedish	●	●	●	●	●	●	●	●	●
German	●	●	●	●	●	●	●	●	●
Spanish	●	●	●	●	●	●	●	●	●

Teaching Qualification (Sweden)

Teaching and Learning in
Higher Education, 2022 Spring,
Lund University
Communicating Science, 2024
Fall, Lund University

Translator Work

天使升华在空中 (Sublimation
Angels), Douban Read

Social

-  GitHub
-  Google Scholar
-  ORCID
-  LinkedIn
-  Lund University

RECENT GRANTS AND AWARDS

WASP Research Stint Abroad grant for visiting ETH Zurich	<i>The Wallenberg AI, Autonomous Systems and Software Program (WASP)</i>	2024
Young Researchers Travel grant	<i>Royal Physiographic Society in Lund</i>	2023 & 26
International Merit Scholarship	<i>Faculty of IT, Monash University</i>	2016

RECENT RESEARCH COMMUNITY SERVICE

Organizing Activities: Equity Committee member for The ACM conference on Human factors in Computing systems (**CHI**). Co-chair of Publicity for The **<Programming>** Conference 2025. Co-chair for the Posters and Demos track of The **<Programming>** Conference 2024. Student volunteer for The **Psychology of Programming Interest Group (PPIG)** 2023.
Reviewing Activities: The ACM conference on Human factors in Computing systems (**CHI**) 2026. The journal of Transactions on Software Engineering and Methodology (**TOSEM**) 2024

RECENT OUTREACH ACTIVITIES

Participation in the Swiss ACM SIGCHI Chapter (SwissCHI) Winter Meeting, Università della Svizzera italiana, Switzerland, 2025
Poster at the Natural and Artificial Cognition Workshop, Lund, Sweden, 2024
Presentation at the PPIG & VL/HCC conference, Liverpool, UK, 2024
Poster at WASP study trip to the Silicon Valley, USA, 2024
Presentation at WARA-Ops Workshop on Data-Driven Research and Operations (hosted by Ericsson), Lund, Sweden, 2024
Presentation on Eye Tracking at a partner company Neo4j, Malmö, Sweden, 2024
Presentation at the ICSE conference, Melbourne, Australia, 2023

RECENT TEACHING

Teaching Assistant (TA), Agile Software Development (ongoing), since 2023 Spring
TA, Scientific Theories for Engineers, 2026 Spring
TA, Software Testing, 2024 and 2025 Fall
TA, Software Development for Large-scale Systems, 2024 and 2025 Fall
TA, Database Technology, 2024 Spring
TA, Object-oriented Modelling and Design, 2021 and 2022 Fall

RECENT SUPERVISION

Bachelor's thesis: Improving Advanced Information Models Using LLMs
Master's thesis on Retrieval-Augmented Generation for Educational Exploration
Bachelor's thesis on Locally Deploying an AI Chatbot to Limited Hardware with Custom Data
Assisting in supervising six research assistants on the Gander (Gazing at Code Review) project

THESES

Peng Kuang. 2026. *Building Gaze-aware Programming Environments (in progress)*. Doctoral dissertation. Lund University, Lund, Sweden.
Peng Kuang. 2024. *Toward Gaze-enabled Programming Tool Assistance*. Licentiate thesis. Lund University, Lund, Sweden.

RECENT PUBLICATIONS

Peng Kuang, Emma Söderberg, April Yi Wang, and Martin Höst. 2025. GazePrinter: Visualizing Expert Gaze to Guide Novices in a New Codebase. *Manuscript under review*.
Peng Kuang, Emma Söderberg, Diederick C. Niehorster, and Martin Höst. 2023. *Toward Gaze-assisted Developer Tools*. In *Proceedings of the 45th International Conference on Software Engineering: New Ideas and Emerging Results (ICSE-NIER)*.
Peng Kuang, Emma Söderberg, and Martin Höst. 2024. *Developers' Perspective on Today's and Tomorrow's Programming Tool Assistance: A Survey*. In *Proceedings of the 10th Edition of the Programming Experience Workshop (PX)*.
Peng Kuang, Emma Söderberg, and Martin Höst. 2024. *Designing A Multi-modal IDE with Developers: An Exploratory Study on Next-generation Programming Tool Assistance*. In *Proceedings of 35th Annual Workshop of The Psychology of Programming Interest Group (PPIG)*.
Peng Kuang, Emma Söderberg, Diederick C. Niehorster, and Martin Höst. 2023. *Applying Machine Learning to Gaze Data in Software Development: A Mapping Study*. In *Proceedings of the International Workshop on Eye Movement in Programming (EMIP)*.
Alan T. McCabe, Moa Björkman*, Joel Engström*, **Peng Kuang**, Emma Söderberg, and Luke Church. 2024. *Ironies of Programming Automation: Exploring the Experience of Code Synthesis Via Large Language Models*. In *Proceedings of the 8th International Conference on the Art, Science, and Engineering of Programming*.
Alan T. McCabe, Emma Söderberg, Luke Church, **Peng Kuang**, 2022. *Visual Cues in Compiler Conversations*. In *Proceedings of the 33rd Annual Workshop of the Psychology of Programming Interest Group Annual Workshop (PPIG)*.
William Saranpää, Theodor Fransson, Hjalmar Rutberg, Emma Söderberg, **Peng Kuang**, Diederick C. Niehorster. 2025. Understanding Developers' Gaze in Code Review. *Work in Progress*.
Kashumi Madampe, John Grundy, Judith Good, Dulaji Hidellaarachchi, Jácome Cunha, Chris Brown, **Peng Kuang**, Reham Al Tamime, Ariful Islam Anik, Advait Sarkar, Wei Zhou, Shawal Khalid, Tommaso Turchi, Shavindra Wickramathilaka, and Yue Jiang. 2025. *Addressing the Agony of Recruitment for Human-centric Computing Studies* In *SIGSOFT Software Engineering Notes* 50, 2 (April 2025), 23–26..